

SUMANTH KUMAR MEESALA

Senior LLM & Agent Application Engineer • AI-Native Full-Stack • Node.js • React • TypeScript

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EXECUTIVE SUMMARY

Sumanth Kumar Meesala is a senior full-stack engineer with 11+ years of production experience across enterprise SaaS, geospatial intelligence, and rapid client product builds. Over the last three years he has converged that engineering depth with a distinct AI-native product engineering practice — shipping LLM-powered features, retrieval systems, and bounded autonomous agents on top of typed JavaScript stacks. He operates Claude Code, Cursor, GitHub Copilot, and Augment Code as primary tools, not novelties, and treats AI agents as engineering teammates with explicit responsibilities, context, and guardrails. He is most effective in environments that demand both rapid 0→1 shipping and production-grade engineering rigour — founding/early-stage AI engineering teams, AI-native product squads inside scaleups, and senior LLM application engineering roles where AI features must be merged with hard system constraints.

WHAT I OPTIMISE FOR

- **Shipping AI features that actually run in production** — not demos. Latency budgets, cost ceilings, graceful failure modes, and observability are part of every LLM design I do.
- **AI-accelerated engineering velocity** — turning Claude Code, Cursor, and similar tools into a force multiplier through context engineering, slash commands, agent guardrails, and team standards (CLAUDE.md / AGENTS.md).
- **Bounded autonomy** — designing agents that do one well-scoped job repeatedly, with auditable side-effects, instead of free-form “do everything” agents that drift in production.
- **Engineering credibility over hype** — my comfort zone is Node.js, React, TypeScript, AWS; AI is layered on, not pasted over.

CORE COMPETENCIES

AI-Native Product Engineering: LLM application architecture, RAG pipelines (chunking, embeddings, retrieval, re-ranking), context engineering, multi-step agentic workflows, multi-agent orchestration, tool / function calling, structured outputs, evals, guardrails, prompt engineering, AI workflow automation

AI Tooling (Daily Driver): Claude Code, Cursor, GitHub Copilot, Augment Code, MCP servers, autonomous coding agents, AI-assisted code review, agent-driven test generation, AI-assisted refactoring at scale

LLM Stack: OpenAI GPT-4 / GPT-4o, Anthropic Claude (Sonnet / Opus), Llama, Hugging Face; LangChain, LangGraph, LlamaIndex, CrewAI; Pinecone, ChromaDB, FAISS, pgvector

Languages: TypeScript, JavaScript (ES6+), Node.js, Python (used for LLM tooling and pipelines), C# (legacy .NET enterprise)

Backend: Node.js, Express, AWS Lambda, Serverless Framework, REST APIs, WebSockets, SSE, JWT, OAuth 2.0, async data pipelines

Frontend: React 18, Redux / Zustand, AngularJS, Tailwind, Material UI, responsive + real-time UIs, design-system thinking

Data: DynamoDB, MongoDB, SQL Server, MySQL; vector stores (Pinecone, Chroma, pgvector, FAISS)

Cloud & DevOps: AWS (Lambda, API Gateway, CloudFormation, Cognito, S3, DynamoDB, EC2, CloudFront), Docker, CI/CD, Git, JIRA

Practices: Rapid prototyping, MVP shipping, system architecture, technical leadership, code review, spec-driven workflows with AI agents, observability for LLM systems

PROFESSIONAL EXPERIENCE

DashAnalysis Pty Ltd — Senior Full-Stack & AI Engineer

Aug 2020 – Present

Lead engineer for AI capabilities on a multi-tenant analytics platform; co-designs architecture and owns production delivery of all AI-driven features.

- Architected and shipped enterprise-grade RAG pipelines combining LangChain and LlamaIndex on a hybrid Node.js / Python services backbone, indexing technical documentation and product data into Pinecone and ChromaDB. Tuned chunking, embedding, retrieval, and re-ranking strategies; ~40% improvement in retrieval efficiency vs. baseline keyword search. Replaced multi-day analyst lookups with sub-minute, source-cited answers.
- Designed and operates a fleet of autonomous AI agents using LangGraph and CrewAI to automate analytics, reporting, and DynamoDB-driven orchestration. Each agent has a narrow role (e.g. fetch → reason → write report → notify), explicit tools, and idempotent side-effects — so they can be retried, audited, and rolled back.
- Integrated Claude (Sonnet / Opus) and GPT-4o into Node.js + AWS Serverless backends. Built a shared LLM client layer with retry, timeout, structured-output enforcement (Zod-style schemas), prompt-caching, and per-tenant rate limits. Hallucination incidents and per-request cost both dropped meaningfully after rollout.
- Productionised AI-assisted engineering as the team's default workflow. Standardised Claude Code, Cursor, and Augment Code; authored shared CLAUDE.md / AGENTS.md context files, custom slash commands, and MCP-style tool integrations; embedded AI agents into the code-review and test-generation loops. Median feature cycle compressed by ~30%.
- Led the React 18 + TypeScript front-end re-architecture, gradually retiring legacy AngularJS modules. Built real-time analytics dashboards over WebSocket streams from the Node.js API and a typed component library used across product surfaces.
- Acts as informal architecture lead for AI features — reviews proposals, owns latency / cost budgets for LLM calls, and is the point person for new model adoption.
- Co-built Moose (beige.tech/solutions/moose) — enterprise product platform on AngularJS front-end + AWS serverless backend (Lambda, CloudFormation, S3, Cognito, DynamoDB). Owned full-stack delivery and infrastructure-as-code rollout; designed auth flows on Cognito, async data pipelines on Lambda, and document/object storage on S3 + DynamoDB.
- Designed, shipped, and operate qrank.it (qrank.it.com) — a 360° employee review application on AngularJS + Node.js + AWS. Integrated LLM to convert structured peer / manager / self feedback into AI-generated, human-readable review reports. Also used as a real-world testbed for rapid AI prototyping patterns: Claude Code-driven development, AI-generated test scaffolding, and quick iteration on production AI behaviour.
- Delivered bespoke multi-step agentic workflows (LangGraph + Node.js) for SMB / enterprise clients integrating with their APIs, internal tools, and data stores. Pattern: bounded autonomy — defined input contracts, fixed tool sets, structured outputs, explicit termination conditions, and reversible side-effects. Days of manual back-office work compressed into minutes.

Blaque Fracture Technologies Pty Ltd — Senior Full-Stack Developer

Jul 2019 – Jul 2020

Senior IC role on a small product team delivering B2B web applications across multiple client engagements.

- Translated business requirements into MERN-stack applications (MongoDB, Express, React, Node.js); owned features end-to-end from spec through to production deployment.
- Designed responsive, state-driven React + Redux interfaces with reusable component patterns; standardised form validation, error handling, and loading states across the product surface.
- Built and maintained REST APIs with JWT-based authentication, role-based access control, and audit logging on sensitive endpoints; wrote MongoDB aggregation pipelines and indexes for read-heavy workloads.
- Integrated third-party services (payment, email, file storage) behind clean adapter layers; participated in technical-design reviews, code review, and release cycles.

Archimedes — Software Developer

Nov 2018 – Feb 2019

Short-term engagement focused on shipping secure customer-facing web flows.

- Built React frontends bridged to Node.js + MongoDB backend services; converted design specs into production-ready, accessible components.

- Implemented OAuth 2.0 authentication and authorization (Google / corporate IdP) with secure session handling, refresh-token rotation, and protected routes.
- Wrote modular Express middleware for request validation, rate limiting, and structured error responses; documented API contracts for client teams.

Blockfreight, Inc. — Software Developer

Aug 2018 – Oct 2018

Product engineering on an early-stage logistics platform.

- Engineered single-page applications using React and Redux integrated with RESTful Node.js / Express endpoints backed by MongoDB.
- Built domain list/detail flows, search, and filtering with focus on perceived performance — skeleton loaders, optimistic updates, and paginated lists for large datasets.
- Authored API services with consistent input validation, error contracts, and logging; improved deployment scripts and developer onboarding documentation.

ContentTerra Software Pvt. Ltd. — Software Developer

Feb 2015 – Jun 2018

Three-year tenure delivering .NET enterprise web applications across multiple client engagements; first professional engineering role and the foundation of my product engineering practice.

- Delivered .NET enterprise web applications using ASP.NET MVC, WCF, Entity Framework, C#, HTML5, and JavaScript across the full SDLC — requirements, design, build, test, release, and support.
- Designed and built data-access layers; optimised SQL Server queries with LINQ, indexed views, and stored procedures to meet enterprise SLA requirements on large transactional datasets.
- Implemented authentication and authorization (Forms / Windows / claims-based) and integrated with third-party SOAP and REST services for upstream and downstream systems.
- Reviewed pull requests, ramped junior team members on the codebase, and maintained engineering standards across release cycles.

PERSONAL PROJECT — CONTENT FACTORY

Solo build · Live in production · 500+ YouTube subscribers · 2 lakh+ video views

End-to-end AI content factory I designed, built, and operate on my own. Automates the full script → TTS → video → publish pipeline for a Telugu astrology YouTube channel and other content surfaces — every step driven by LLM agents and open-source models, glued together by typed full-stack code.

- **Stack:** Next.js 16 (App Router) + TypeScript · Tailwind / DaisyUI / Framer Motion · Remotion (programmatic video rendering) · Claude Agent SDK (script generation) · Edge TTS + custom Parler-TTS server (Hugging Face open-source model) · PostgreSQL · googleapis (YouTube + Instagram OAuth 2.0) · @react-pdf/renderer · Cheerio + RSS Parser (trend scraping) · astronomy-engine + mhah-panchang (Vedic calculations).
- **What it does:** Multi-channel content generation (Astrology, Current Affairs, Trending) with channel-specific AI personas; persona-driven script generation via Claude; multi-language TTS in English, Hindi, Kannada, Malayalam, Tamil, Telugu; word-level animated subtitles synced to TTS; batch render of all 12 zodiac signs in one run; auto-publish to YouTube and Instagram with auto-generated metadata, playlists, and scheduling.
- **Extras:** Trend discovery (Google Trends / YouTube / Twitter / News) feeding AI content recommendations; festival calendar (panchangam) with templated content for Hindu festivals; Vedic Jatakam birth-chart generator (lagna, dasa, dosha, PDF export).
- **Outcome:** Live YouTube channel at 500+ subscribers and 2 lakh+ cumulative video views — running unattended end-to-end. Acts as a portfolio of AI-native engineering fundamentals: agentic LLM orchestration, open-source model integration (Hugging Face), programmatic video pipelines, Postgres-backed workflows, and OAuth-based platform integrations.

ENGINEERING PHILOSOPHY

- AI features ship like every other feature — with tests, observability, and fall-back behaviour.
- Treat agents as engineers with narrow job descriptions: define their tools, inputs, outputs, and stop conditions explicitly.

- Context is a first-class artefact — CLAUDE.md / AGENTS.md, system prompts, and tool docs deserve the same review rigour as code.
- Bias to shipping. The MVP that exists in production teaches more than the perfect MVP that doesn't.
- Respect engineering fundamentals — type safety, observability, and clear ownership beat any model upgrade.

KEY TECHNICAL ACHIEVEMENTS

- **Cut feature delivery time ~30%** by standardising Claude Code, Cursor, and Augment Code as the team's default workflow with shared context files, slash commands, and agent guardrails.
- **Improved retrieval efficiency ~40%** for an enterprise documentation corpus by replacing keyword search with structure-aware chunking + hybrid LangChain/LlamaIndex retrieval over Pinecone.
- **Compressed multi-day analyst tasks to sub-minute, citation-backed answers** with a production RAG service used in real reporting workflows.
- **Replaced manual SOP-driven workflows with bounded autonomous agents** for SMB and enterprise clients — multi-step LangGraph + Node.js agents with auditable, reversible side-effects.
- **Stabilised hallucination rate and per-request cost** in production by introducing a shared LLM client layer with retry, structured-output enforcement, prompt caching, and per-tenant rate limits.
- **Owns and operates two independent products end-to-end** (Moose @ Beige.tech, qrank.it) on Node.js + React + AWS with LLM-assisted features.

WHAT I'M LOOKING FOR

Senior LLM / Agent application engineering, AI-native full-stack, founding AI engineer, or staff/principal-track roles inside India's AI-native startups, AI-first SaaS scaleups, or distributed teams hiring senior engineers in India / remote. Most interested in: 0→1 product surfaces with a credible AI thesis; teams that treat AI engineering as production engineering (evals, observability, latency budgets); founders / EMs who want an early hire who can ship product, design system architecture, and raise the bar on AI-assisted engineering practices for the rest of the team.

EDUCATION & LINKS

Master of Information Technology (Data Analytics) — Deakin University, Australia · Jul 2018 – Jul 2020 · WAM 81.25

B.Tech, Computer Science Engineering — CVR College of Engineering · Aug 2011 – May 2015

Portfolio: beige.tech/solutions/moose · qrank.it.com · github.com/sumanth-kumar-measala · linkedin.com/in/sumanthkumarmeasala